16.1

Logit: $P(AUTO = 1) = \Lambda(-0.2376 + 0.5311 \cdot DTIME)$, $\Lambda(z) = 1/(1 + e^{-z})$; *DTIME* in 10-minute units.

a. $z = -0.2376 + 0.5311(1) = 0.2935$.

$$P = \Lambda(0.2935) = 1/(1 + e^{-0.2935}) = 0.5728$$

Estimated probability of choosing the automobile ≈ **0.573**.

b. Probit index $-0.0644 + 0.3000 \cdot DTIME$; at $DTIME = 1$, $z = 0.2356$, so $P = \Phi(0.2356) \approx$ **0.5931**. The probit (≈0.593) and logit (≈0.573) estimates are very close — both predict a 57–59% probability — illustrating that logit and probit give very similar fitted probabilities.

c. Logit $ME = \tilde{\gamma}_2 \cdot \Lambda(z) \cdot [1 - \Lambda(z)]$. At $DTIME = 3$ (30 min longer by bus): $z = 1.3557$, $\Lambda = 0.7951$, $ME = 0.5311 \times 0.7951 \times 0.2049 =$ **0.0865** (≈8.65 pp). The LPM gives a constant ME = **0.0703** (≈7.03 pp) for any *DTIME*. Close in size, but the LPM effect is constant while the logit effect varies and is largest in the middle of the probability range.

d. 10-min decrease for someone 50 min longer by driving ⇒ $DTIME = -5$. Logit: $z = -2.8931$, $\Lambda = 0.0524$, $ME = 0.5311 \times 0.0524 \times 0.9476 =$ **0.0264** (≈2.64 pp). Probit: $ME = \tilde{\gamma}_2 \cdot \phi(z) = 0.3000 \times \phi(-1.5644) = 0.3000 \times 0.1170 =$ **0.0351** (≈3.51 pp). Both are small because the probability is near zero there (flat part of the cdf); the two estimates are again of similar magnitude.

16.4

Logit: $P(y = 1) = \Lambda(-1.836 + 3.021 \cdot x)$; maximized log-likelihood = −1.612.

a. $z = -1.836 + 3.021(1.5) = 2.6955$.

$$P(y = 1) = \Lambda(2.6955) = 0.9368$$

Estimated probability ≈ **0.937**.

b. 0.9368 > 0.5 ⇒ predict $\hat{y} = 1$. Actual first observation is $y = 1$, so the prediction is **correct**.

c. Data $(x, y) = (1.5, 1), (0.6, 0), (0.7, 1)$. With $\gamma_1 = -1$, $\gamma_2 = 2$:

Obs 1: $z = 2.0$, $\Lambda = 0.8808$ ($y = 1 \rightarrow 0.8808$)

Obs 2: $z = 0.2$, $\Lambda = 0.5498$ ($y = 0 \rightarrow 1 - \Lambda = 0.4502$)

Obs 3: $z = 0.4$, $\Lambda = 0.5987$ ($y = 1 \rightarrow 0.5987$)

$$L = 0.8808 \times 0.4502 \times 0.5987 = 0.2374$$

At the MLEs, $L^* = e^{-1.612} = 0.1995$. So L (0.2374) at $\gamma = (-1, 2)$ is **larger** than L* (0.1995) at the reported MLEs — showing how the likelihood is evaluated and compared across parameter values.

d. True. Likelihood (16.14) is a product of factors $\Lambda^y \cdot (1 - \Lambda)^{1-y}$. Each factor lies strictly in (0, 1), so any finite product is also in (0, 1). Hence the likelihood is always between zero and one.

e. True. Log-likelihood (16.15) sums $y \cdot \ln \Lambda + (1 - y) \cdot \ln(1 - \Lambda)$. Since $0 < \Lambda < 1$, both $\ln \Lambda$ and $\ln(1 - \Lambda)$ are negative, so every term — and the total — is negative. (Equivalently, log of a number in (0, 1) is negative.)

16.18

a. Signs all reasonable. **INSUR (−0.482, highly sig)** is the strongest predictor — insured mortgages are far less likely to be delinquent. **RATE (+0.034)** and **ARM (+0.128)** raise risk; **LVR (+0.0016)** raises it (more leverage); **CREDIT (−0.00044)** lowers it; **REF (−0.059)** and **TERM (−0.013)** negative; AMOUNT (+0.024) positive but insignificant. All intuitive.

b. Logit gives the **same signs** on every coefficient. Significance is mostly the same but weaker for a few: INSUR, RATE, REF, TERM, ARM significant in both; LVR ($p = 0.099$), CREDIT ($p = 0.065$) and AMOUNT ($p = 0.056$) are significant/borderline in the LPM but only marginal in logit. Both agree on direction and on the dominant role of INSUR.

c. Obs 500: LPM = 0.183, logit = 0.134 → both classify as **not delinquent**. Obs 1000: LPM = 0.579, logit = 0.627 → both classify as **delinquent**. Same classification from both models, with slightly different probabilities.

d. AMOUNT = 2.5, LVR = 80, RATE = 8, TERM = 30, indicators = 0:

CREDIT 500: LPM 0.553, logit 0.497

CREDIT 600: LPM 0.509, logit 0.409

CREDIT 700: LPM 0.465, logit 0.326

Both show probability **falling as CREDIT rises**. LPM falls in equal steps of 0.044 per 100 points (constant slope); logit falls by larger amounts at low credit (nonlinear S-shape). The CREDIT histogram clusters at higher scores, where the two models diverge more.

e. Marginal effect of CREDIT (per 1 point):

CREDIT 500: LPM −0.000442, logit −0.000891

CREDIT 600: LPM −0.000442, logit −0.000861

CREDIT 700: LPM −0.000442, logit −0.000783

LPM ME is **constant** (= the coefficient); logit ME **varies**, largest where fitted p is near 0.5 (low credit), shrinking as credit rises. Interpretation: +1 credit point lowers delinquency by ~0.0004–0.0009; +100 points lowers it by about 4–9 percentage points.

f. CREDIT = 600, others as (d). LVR 20: LPM 0.412, logit 0.235. LVR 80: LPM 0.509, logit 0.409. Both show risk **rising with leverage**. LPM rises +0.097 (constant slope × 60); logit rises more in relative terms (+0.174) through the steep middle of the curve. The LVR histogram concentrates near 80, where delinquency risk is highest.

g. Correct predictions (threshold 0.5): LPM = **85.8%** (858/1000), logit = **85.4%** (854/1000) — essentially tied. LPM catches slightly more delinquents (131 vs 118 true positives); logit catches slightly more non-delinquents (736 vs 727). Both are dominated by the large non-delinquent class.

h. No — 0.5 is **not the best rule**. Trained on obs 1–500 and applied to 501–1000, the 0.5 rule misclassifies 60 cases (12.0% error), and 0.5 is near the error-minimizing point only because the delinquency base rate is just 13.4% (predicting "nobody delinquent" already gives ~13% error). The real issue: **overall error is the wrong objective for a lender** — approving a loan that defaults costs far more than declining a good applicant. A better rule uses a **lower threshold** (decline when p exceeds ~0.2–0.3) to screen out more risky applicants, and chooses the threshold to minimize expected **cost** (weighting the two error types by their dollar consequences) rather than the raw misclassification rate.